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Business Case 4 Group 12

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1. Introduction

The objective of this business case is to design an Early Warning System for detecting abnormal financial market conditions. In financial markets, the distinction between *normal* and *abnormal* regimes is economically meaningful because market crises are typically associated with risk-off dynamics: higher volatility, stronger cross-asset correlations, wider credit spreads, falling risky assets, and a general flight to safety. For applications such as robo-advisory, portfolio monitoring, trading platforms, and risk-management dashboards, detecting these transitions early is valuable because it allows the system to react before the largest part of the drawdown has occurred.

The goal is not to predict crises in a strict forecasting sense. The problem is better interpreted as a nowcasting task: the model should recognize the early emergence of market stress using the information currently available in a large cross-asset dataset. This means that, in our case, a false negative, which is failing to detect an actual stress regime, is more costly than a false positive, because missing a crisis may expose the portfolio or the client to severe losses. For this reason, the project prioritizes early detection and recall-oriented evaluation, while still controlling the number of false alarms.

2. Dataset and Feature Engineering

The dataset contains weekly observations of a broad set of financial variables covering equity indices, bond indices, government yields, interest rates, currencies, commodities, volatility indicators, and macro-financial leading indicators. The target variable identifies whether each week belongs to a normal or abnormal market regime. This structure naturally frames the exercise as a binary classification problem, but in reality it is closer to regime detection: the abnormal class represents periods in which market behaviour deviates materially from its usual risk-on configuration.

Before modelling, the raw market variables must be transformed into stable and comparable features. This step is central because many financial time series are non-stationary in levels. Equity indices, bond indices, currencies, and commodities are therefore transformed mainly through log-returns, while yields and interest rates are treated through first differences. Variables such as implied volatility or macro surprise indicators can instead be retained in levels when their economic meaning is already stationary or directly informative. The purpose of this transformation is to make the model learn changes in market conditions rather than long-run trends in asset prices.

The preprocessing phase also has a financial rationale. During crises, abnormality is rarely visible in one isolated asset. It usually appears as a joint configuration: equity losses, volatility spikes, spread widening, safe-haven flows, and changes in rates. Therefore, the final feature space is designed to preserve cross-market information and allow the models to detect multivariate deviations from normal behaviour.

3. Gaussian Baseline and Multivariate Anomaly Detection

The first modelling step is a Gaussian baseline based on the multivariate normal distribution. The core assumption is that normal market behaviour can be approximated by a joint distribution of the transformed features. The model is trained using normal observations only, so that it learns the structure of regular market conditions without being directly influenced by crisis periods.

For each observation, the model computes an anomaly score based on its distance from the estimated normal distribution. In practice, this is equivalent to assigning a high Mahalanobis distance to observations that are far from the centre of normal market behaviour. An observation is then classified as abnormal when its score exceeds a threshold. The threshold is selected on the validation set by optimizing an evaluation metric (F2 score) that gives greater importance to detecting true crises.

This baseline is useful for two reasons. First, it is transparent: a high anomaly score has a clear interpretation as a deviation from the historical normal regime. Second, it provides a benchmark against which more complex models can be compared. However, the baseline also has limitations. Financial variables are correlated, high-dimensional, and often affected by unstable covariance estimates. For this reason, the baseline is improved through covariance regularization.

4. Model Refinements and Domain-Informed Design

A first refinement consists of replacing the empirical covariance matrix with a Ledoit–Wolf shrinkage estimator. The purpose is to make the covariance estimate more stable in a high-dimensional setting, reducing the risk that the anomaly score is driven by estimation noise. Robust scaling is also tested to reduce the influence of extreme observations on the estimation of location and dispersion.

A further improvement is obtained by introducing financial domain knowledge. Instead of treating all variables as a single homogeneous block, the features are grouped by asset class: equity, rates, foreign exchange, commodities, bond indices, and leading indicators. A separate anomaly detector is fitted to each group, and the resulting scores are combined. This structure reflects the fact that a crisis can emerge through different channels. For example, an equity-led sell-off, a rates shock, or a credit-market stress episode may have different signatures, even if they all correspond to risk-off conditions.

The domain-informed model also incorporates the role of volatility. Since a genuine risk-off regime is usually associated with a volatility spike, a volatility-based filter can be used to reduce false alarms. This is really helpful since not every statistical deviation is economically relevant, but a deviation occurring together with elevated volatility is more likely to represent systemic stress.

5. Interpretability and Crisis Characterization

Interpretability is a key requirement for an Early Warning System. A risk manager does not only need to know that the model is raising an alarm; they also need to understand why. For the Gaussian-based models, the anomaly score can be decomposed into feature-level contributions. This allows the system to identify which variables are pushing a given observation away from normal behaviour.

This interpretation is similar in spirit to feature attribution methods used in machine learning. If volatility, equity indices, credit-sensitive bond indices, or safe-haven currencies dominate the contribution, the alert can be interpreted consistently with risk-off theory. This is important because an Early Warning System should not behave like a black box: its output must be economically explainable and operationally usable.

Then, the detected abnormal weeks are studied by mapping them onto known crisis periods and by clustering them into groups. This step recognizes that crises are not all identical. Some may be equity-driven, others may originate from credit markets, commodities, or rates. Clustering the anomalous observations helps assign a financial label to different types of stress episodes and provides a richer interpretation than a simple normal/abnormal classification.

6. Comparison with Alternative Models

Several alternative models are considered to evaluate whether more flexible techniques can improve the detection of abnormal regimes. Supervised methods such as Random Forests, Support Vector Machines with radial kernels, and gradient boosting models exploit the labelled target directly. These models can capture nonlinear interactions between variables and may identify complex decision boundaries that a Gaussian detector cannot represent. For all these models we implemented automatic hyperparameter selection through OPTUNA and

the optimized supervised models ended up being the ones with the best performance.

A sequence-based approach is also considered. An LSTM autoencoder is trained exclusively on normal observations and learns to compress and reconstruct temporal windows of 30 trading days. At inference time, observations that the model fails to reconstruct accurately, measured by the mean squared error on the final timestep, are flagged as anomalies. Unlike point-in-time detectors, this architecture explicitly captures the sequential structure of market dynamics, making it sensitive to abnormal patterns that only manifest over several consecutive periods.

Hybrid approaches are also explored. For example, a supervised classifier can be combined with an unsupervised outlier detector such as COPOD, so that the final decision reflects both the probability of belonging to the abnormal class and the degree of statistical extremeness. Semi-supervised PCA is another relevant approach: it learns the structure of normal data and flags observations that are poorly reconstructed. Finally, fully unsupervised methods such as Local Outlier Factor and Isolation Forest are tested as additional benchmarks, even though they do not outperform supervised methods.

The models are compared using metrics that reflect the asymmetric costs of the business problem. Precision measures the reliability of the alarms, while recall measures the ability to detect actual stress periods. The F2 score is particularly appropriate because it assigns more weight to recall than to precision, which is coherent with the objective of avoiding missed crises. ROC-AUC is used as a threshold-independent measure of ranking quality, while the confusion matrix provides a direct view of false positives and false negatives.

7. Financial Interpretation and Conclusion

An Early Warning System transforms a large cross-asset information set into an operational risk signal. The signal can support defensive portfolio allocation, risk-budget reduction, client monitoring, or systematic strategy design. In a wealth-management context, it can help protect clients from severe drawdowns by identifying periods in which the market environment becomes structurally different from normal conditions. In an investment context, the same signal can be used to dynamically adjust exposure between aggressive and defensive regimes. The most important methodological insight is that anomaly detection is not simply a statistical exercise. The definition of abnormality must be aligned with the business objective. In this case, the relevant anomaly is not any unusual observation, but an unusual observation that is financially consistent with stress. Therefore, the best Early Warning System is not necessarily the most complex model, but the one that balances early detection, false-alarm control, robustness, and interpretability.

Overall, the project builds a complete framework for detecting financial stress regimes: data transformation, baseline anomaly detection, covariance stabilization, domain-informed modelling, interpretability, crisis characterization, and comparison with supervised, semi-supervised, and unsupervised alternatives. The resulting methodology provides a coherent foundation for a risk-monitoring tool capable of recognizing the transition from risk-on to risk-off market conditions.